

Assesment for Hotel Management System

7-03-2026

Technology: Spring Boot + Spring Data JPA + Oracle Database

Objective

Develop a **Hotel Management System** using **Spring Boot and Spring Data JPA** that manages hotel operations.

The system must follow a **3-layer architecture**:

- **Controller Layer** – Handles HTTP requests
- **Service Layer** – Business logic
- **Repository Layer** – Database access using Spring Data JPA

Students must implement **CRUD operations** for two modules.

Case Study

A hotel wants to **digitize its management system** to handle rooms and customer bookings efficiently.

The system should allow the hotel administrator to:

1. Manage hotel rooms
2. Manage customer bookings
3. Track which customer booked which room
4. Update booking status
5. Delete bookings if cancelled

The application must store data in an **Oracle database**.

Modules

Module 1: Room Management

The hotel has different types of rooms such as **Single, Double, Deluxe, Suite**.

Room Table Fields

Field	Type	Description
roomId	Number	Primary Key
roomNumber	Varchar	Unique room number
roomType	Varchar	Single / Double / Deluxe
price	Number	Price per day
status	Varchar	Available / Occupied

Required CRUD Operations

Students must implement:

1. Create Room

POST /rooms

2. Get All Rooms

GET /rooms

3. Get Room By ID

GET /rooms/{id}

4. Update Room

PUT /rooms/{id}

5. Delete Room

DELETE /rooms/{id}

Module 2: Booking Management

Customers can book rooms in the hotel.

Booking Table Fields

Field	Type	Description
bookingId	Number	Primary Key
customerName	Varchar	Customer name
customerPhone	Varchar	Phone number
checkInDate	Date	Check-in date
checkOutDate	Date	Check-out date
roomId	Number	Foreign key from Room

Required CRUD Operations

1. Create Booking

POST /bookings

2. Get All Bookings

GET /bookings

3. Get Booking By ID

GET /bookings/{id}

4. Update Booking

PUT /bookings/{id}

5. Cancel Booking

DELETE /bookings/{id}

Application Architecture

Students must follow this **project structure**:

```
com.hotelmanagement
├── controller
│   ├── RoomController
│   └── BookingController
└── service
```

```
RoomService
BookingService

service.impl
RoomServiceImpl
BookingServiceImpl

repository
RoomRepository
BookingRepository

entity
Room
Booking

HotelManagementApplication
```

Oracle Database Tables

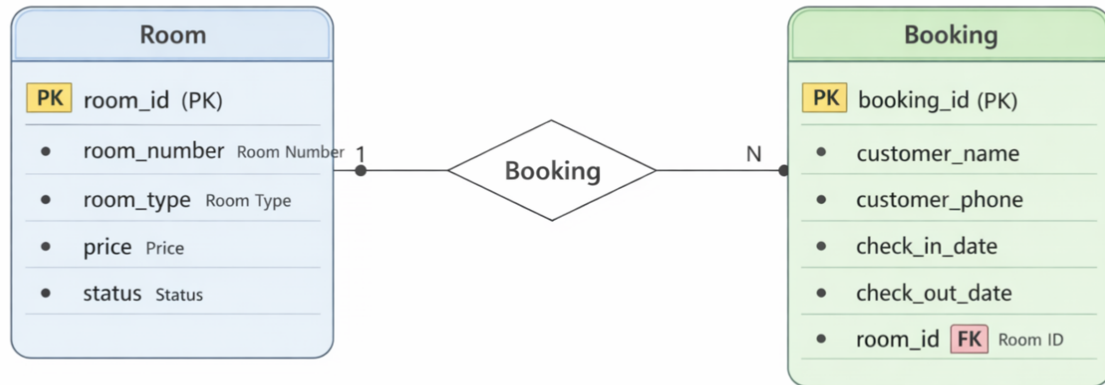
Room Table

```
CREATE TABLE ROOM (
    ROOM_ID NUMBER PRIMARY KEY,
    ROOM_NUMBER VARCHAR2(20),
    ROOM_TYPE VARCHAR2(20),
    PRICE NUMBER,
    STATUS VARCHAR2(20)
);
```

Booking Table

```
CREATE TABLE BOOKING (
    BOOKING_ID NUMBER PRIMARY KEY,
    CUSTOMER_NAME VARCHAR2(100),
    CUSTOMER_PHONE VARCHAR2(20),
    CHECKIN_DATE DATE,
    CHECKOUT_DATE DATE,
    ROOM_ID NUMBER,
    CONSTRAINT FK_ROOM
    FOREIGN KEY (ROOM_ID)
    REFERENCES ROOM(ROOM_ID)
);
```

Hotel Management System ER Diagram



PK - Primary Key

FK - Foreign Key

Expected APIs

Room APIs

Method	API	Description
POST	/rooms	Add room
GET	/rooms	Get all rooms
GET	/rooms/{id}	Get room by ID
PUT	/rooms/{id}	Update room
DELETE	/rooms/{id}	Delete room

Booking APIs

Method	API	Description
POST	/bookings	Create booking
GET	/bookings	Get all bookings
GET	/bookings/{id}	Get booking
PUT	/bookings/{id}	Update booking
DELETE	/bookings/{id}	Cancel booking